

CHAPTER 8

Genomics of pediatric cancer

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8.1 Introduction

Cancer is defined as a genetic disease in any organ or tissue of the human body; in cancer, cells start growing and are uncontrollable and exceed beyond their usual boundaries to invade adjoining parts of any of the human body or other organisms. Both malignant tumors and neoplasms were considered to be the alternative names for cancer (Siegel et al., 2024). Although studies have updated the definition of cancer as “Cancer is a disease of uncontrolled proliferation by transformed cells subject to evolution by natural selection,” cancers are caused by changes in single cells with single to multiple genes existing. Furthermore, these cells will be converted as tumors (Brown et al., 2023). Around 10% of germline mutations cause noninherited cancers (Gharehzadehshirazi et al., 2023). Cancer is recognized as a global health issue in human beings of all ages (Siegel et al., 2023). The development of cancer in adults and children will be varying based on the blowout and treatment of the disease. However, all forms of cancer will not spread into the human body of children. Childhood cancer is defined as any type of cancer that occurs during birth (i.e., day 1) until the age of 14 years (Anderson et al., 2000). The most common cancers affecting children are different forms of leukemia, lymphoma, numerous tumors like Wilms tumor, spinal cord and brain tumor, neuroblastoma/retinoblastoma, and bone and soft tissue cancers which are well-known as pediatric cancer (Stiller, 2004). Every year, 0.4 million cases of different cancers are diagnosed globally and 1/285 cases of cancers are diagnosed before the completion of their adolescence. In this chapter, patients aging 0–14 years are categorized as children and those aging 15–19 years are categorized as adults. Around 44% of cancer cases in children were never diagnosed, and if diagnosed, it takes only 80 seconds per case in the world (Bray et al., 2024). In developing countries, the prevalence of cancer in children was found to be 3%–10% and 1% in developed countries. The mortality rate was between 4% and 5% in developing countries and 1% in developed countries. Worldwide, around 0.16 million new cases of cancer were diagnosed in children <15 years of age with 90,000 leading to death (Badar et al., 2009). Childhood cancer is growing rapidly in the present era. The occurrence of pediatric or childhood cancer is expanding in high-income countries. Such cancer is considered to be traumatic for both

the parents and the affected child (Liu et al., 2023). In high-income countries, 80% of childhood cancers were cured compared with only 30% in low- and middle-income countries (Lam et al., 2019; Magrath et al., 2013). Based on WHO reports, every year, around 0.4 million children and adolescents between 0 and 19 years of the age develop different forms of cancers such as brain cancers, lymphoma, and leukemia and various forms of solid tumors such as Wilms tumors and neuroblastoma. Generic medications are the main treatment option for childhood cancers. Alternative treatments, such as radiotherapy and surgeries, are also being used for curing cancer (World Health Organization, 2021). During the last five decades, childhood cancer has been less common. The prevalence of several types of pediatric cancers was determined to be around 15% worldwide in the 1970s, while currently, it is between 5% and 7% among the global pediatric population (Wellbrock et al., 2023). The progression of cancer starts in the human body when cells undergo human genetic amendments, and then, it progresses into a tumor, which further invades and affects other human body parts, which then results to death if untreated. The actual reason for the development of childhood cancer is not accurately determined, but majority of the studies have cited lifestyle and environmental factors to be the cause (World Health Organization, 2021). The analysis of recent distribution and temporal trends of childhood cancer may be significant in the formulation and implementation of preventive programs in different nations, as resource planning and allocation are important to lessen the burden of disease caused by cancer (Huang et al., 2023).

Cancer is considered as one of the top 10 diseases in children. Globally, in 2018, around 0.2 million new cases and 70,000 cancer deaths were recorded in children between 0 and 14 years of age (Malvezzi et al., 2021). Childhood cancer is known to be a multifactorial etiology whose genetic basis is incomplete with extensive involvement of the immune system and moderated by exposure to environmental factors. Currently, preventive measures are ineffective against childhood cancers (Salas et al., 2017). The individuals diagnosed with cancer during childhood will be facing health-related life-threatening complications (Table 8.1) in their mid-40s specifically in one-third of the population. In the future, these conditions can cause premature death (Bhatia et al., 2023). The survival rate of pediatric cancer has been increased from 58% to 85% in 1975–2022 (Siegel et al., 2022). Childhood cancers were found to be histologically very diverse, whereas adult cancers were found to be carcinomas (Steliarova-Foucher et al., 2017). Children who undergo cancer treatment with surgery sequelae, radiotherapy, and cytotoxic chemotherapy are found to develop mental disabilities, secondary cancers, and organ toxicities (Kopp et al., 2012). The noncombination of rare variants and non-Mendelian inheritances which include maternal genetic effects or de novo germline mutations can contribute toward cancer in children (Kuhlen et al., 2019). Cancer patients of different age groups such as pediatric/children, adolescents, and young adults present with a heterogeneous group of diseases which is broadly classified as brain

Table 8.1 Common sequelae of treatment for childhood cancer (Bhatia et al., 2023).

Human body system	Possible outcomes	Exposures for medical therapies
Central nervous system	Impairment of neurocognitive function	Cranial radiation and intrathecal methotrexate
Endocrine	Growth hormone deficiency	Cranial radiation and complete body irradiation
	Central adrenal insufficiency/hypopituitarism/gonadotropin deficiency	Cranial radiation
	Hypothyroidism	Neck radiation and total body irradiation
	Diabetes	Abdominal radiation and total body irradiation
Cardiac	Cardiomyopathy	Chest radiation
	A bunch of cardiovascular diseases	Chest irradiation
Pulmonary	Restrictive pulmonary disease	Chest radiation, bleomycin, pulmonary surgery, and busulfan
Musculoskeletal	Decreased bone mineral density	Corticosteroids, allogenic hematopoietic stem cell transplantation
	Osteonecrosis	Corticosteroids, allogenic hematopoietic stem cell transplantation
	Muscular atrophy, skeletal hypoplasia, scoliosis & kyphosis	Radiation
Neurologic	Peripheral neuropathies	Vinca alkaloids, platinum-based chemotherapy
Reproductive	Primary gonadal insufficiency, ovarian hormone deficiency, and premature ovarian failure	Pelvic/testicular radiations, alkylating agent chemotherapy, and total body irradiation
	Reduced fertility, infertility, and shortened lifetime period of fertility	Pelvic/testicular radiations, alkylating agent chemotherapy, and total body irradiation
Subsequent neoplasms	Basal cell carcinoma	Any type of radiation
	Thyroid cancer	Heat and neck radiations
	Breast cancer	Chest radiation, anthracycline and alkylating agent chemotherapies

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Table 8.1 (Continued)

Human body system	Possible outcomes	Exposures for medical therapies
	Colorectal cancer	Abdominal/pelvic radiations and Complete body irradiation
	Glioma	Cranial radiation
	Meningioma	Cranial radiation
	Sarcoma	Anthracycline chemotherapy, bone radiation/soft tissue

Was completely adopted from Bhatia et al (Bhatia et al., 2023) studies as this is a chapter.

tumors, leukemias, and noncentral nervous system (CNS) solid tumors. These subgroups comprise several clinically different entities, most of which have inadequate molecular characterization (Sweet-Cordero & Biegel, 2019). The etiology of childhood cancers has not been updated to make it understandable for everyone globally. However, previously twin studies have underlined the relevance of environmental factors, while inherited genetic factors make a relatively little contribution to the growth of cancer disease (Lichtenstein et al., 2000; Luke et al., 2005). The interplay between external and hereditary cancer risk factors is the primary cause of genetic and epigenetic abnormalities in human cells (Ning et al., 2014). Children's cancer is most likely caused by prenatal and congenital causes; at least 10% of pediatric cancer patients have germline mutations in genes that predispose them to the disease (Pavlidis et al., 2020). Acute lymphoblastic leukemia (ALL) is associated with the TP53 gene, and germline mutations in SUFU and PTCH1 have been identified as risk factors for baby medulloblastoma (Brodeur et al., 2017; Pui et al., 2015; Waszak et al., 2018). The updated report of the International Agency for Research of Cancer showed considerable variations via age, sex, and geographic regions (Alonso-Luna et al., 2023). The organs involved in cancer and leukemia are composed of numerous histologic types, each of them with its own frequency and demographic trends. The distribution of cancers by anatomic site indicates the combined frequencies of numerous types of cancers, rather than the incidence of a specific type of cancer. The different forms of bone cancers such as chondrosarcoma, osteosarcoma, and Ewing sarcoma differ substantially epidemiologically from one another (Miller et al., 1995).

Pediatric cancers are a diverse set of illnesses caused by a variety of environmental variables, as well as germline and somatic genetic variants. Documenting germline mutations that increase a child's risk of developing cancer could be useful for both therapeutic and research purposes (Zimmerman et al., 2015). Rare diseases make it difficult to enroll a significant number of individuals at any single institution and geographical area, requiring multiinstitutional collaboration (Filbin & Monje, 2019). From the public health point of view, the treatment for childhood cancer was found to

be challenging for the oncologists, the patients, and their caring parents (Kaatsch, 2010). The epidemiological association studies which involve germline genetic variants and precise childhood tumors could suggest genes which might be involved in predilection to precise tumors. Studies of single nucleotide polymorphisms (SNPs) which exist in the germline have notably increased understanding about the molecular basis of Wilms tumor and retinoblastoma (Aplan & Khan, 2011). Genetic syndromes have different levels of association in childhood cancers (Pizzo & Poplack, 2015), that is, Li-Fraumeni syndrome is highly associated with early onset of osteosarcoma (Ognjanovic et al., 2012). Revolutionary advances in SGS analysis such as the next-generation sequencing (NGS) technique combined with rapidly increasing growth in both stem cells and chromatin have escorted in a novel way our understanding of the molecular biology among the pediatric cancers (Gröbner et al., 2018; Ma et al., 2018). A single case report may suggest coincidence rather than causation when it comes to other genetic disorders and childhood cancer. It may be indicative of the pathway's importance in tumor development if many disorders cooccurring with a single cancer type affect genes in a similar pathway (Zimmerman et al., 2015). Worldwide childhood cancer is momentous but moderately neglected due to morbidity and mortality (Bhakta et al., 2019; Pritchard-Jones et al., 2013). Childhood cancers constitute a heterogeneous cancer group in which adult-type carcinomas, which form the basis of the ICD-10 classification system's definition, are uncommon. The International Classification of Childhood Cancers (ICCC-3) and the World Health Organization's third revision of the International Classification of Diseases for Oncology (ICD-03) categorize additional cancer groups into 12 primary diagnostic groups, which correspond to the various types of cancers that affect children (Fritz, 2000; Johnston et al., 2021; Steliarova-Foucher et al., 2005).

Childhood cancers are always fatal without any early diagnosis, as well as treatment (Mullen et al., 2021). Currently, there are no effective population-based cancer screening programs or lifestyle risk reduction methods for enhancing childhood cancer survivability (Bhakta et al., 2019). Treating a child with cancer is a contemporary medicine success story because many diseases that were thought to be incurable now have successful regimens found (Gatta et al., 2009). Over five decades, multidisciplinary, centralized treatment and excellent supporting care have transformed the survival statistics from childhood cancer at an astounding rate (Gatta et al., 2009). The mortality rate of cancer among children and adolescents (0–20 years old) was found to be 50% in 1975–2000, and more specifically, the mortality rate was found to be 5.1% and 2.1%/100,000 children in 1975 and 2020 (Siegel et al., 2023). In the 1970s, 58% of children and 68% of adolescents diagnosed with cancer survived (Erdmann et al., 2021). The half-year survival frequency was 83.2% in a 12-month-old baby, 87.8% in 1–4 years old, 85.7% in 5–9 years old, 85.5% in 10–14 years of age, and 87.3% in teenage groups among children, according to studies conducted between 2013 and 2019. The 5-year

survival rate was found to be increased for ALL from 57% to 92.1% between 1975 and 2019 and that for non-Hodkin lymphoma from 43% to 91.4% between 1975 and 2019 (Siegel et al., 2022). Among glioma, diffuse intrinsic pontine glioma was known to be a rare type of brain tumor in which 50% of the children survived for less than 365 days and only 10% of them were alive for a couple of years (Hargrave et al., 2006). The 5-year survival rate of soft tissue sarcoma was 65.1% in diagnosed rhabdomyosarcoma and 80.8% in diagnosed peripheral nerve sheath tumors/fibroblastoma (Chhikara & Parang, 2023).

8.2 Prevalence of childhood cancers

Childhood cancer varies from adult cancers based on biology, presentation, response to treatment, and epidemiology (de Paula Silva et al., 2024). Based on WHO report in 2022, 0.27 million children and adolescents were diagnosed with cancer and approximately 0.1 million children died because of cancer. The prevalence of cancers in children and adolescence is rare. A total of 15,190 children/adolescents between 0 and 19 years of age were diagnosed with cancer in 2023, and a 10% of them, that is, 1590, have passed away in the United States. In Children between 0 and 14 years of the age, 9910 were diagnosed with and 1040 died due to cancer. When it comes to adolescents, that is, 15–19 years of age, 5280 of them were diagnosed with and 550 died because of this disease (Clancy, 2023). The global prevalence of childhood and adolescent cancers differs by race and ethnicity. The incidence of leukemia in children was found to be high in American Indians/Hispanic and Native Alaskans in 2016–2020. Meanwhile, the occurrence rates of brain cancer and tumors of other nervous system were high in Alaskan/White and American Indian children (Yao et al., 2024). The prevalence of childhood cancer in Scotland has increased from 120 to 161 cases per million children between the years 1983 and 2007 (Bhattacharya et al., 2014). An epidemiological study conducted by Ren et al. (2022) on children aged 0 to 5 years revealed a reduction in the prevalence of childhood cancers during the past 30 years. In 1990–2019, there was a drop in the incidence and prevalence of childhood cancers, from 7.2% to 2.2% and 12.6% to 3.4%, respectively. Additionally, the overall death toll dropped to 47.8%. Testicular cancer cases increased while those of liver cancer, leukemia, and Hodgkin's lymphoma decreased (Ren et al., 2022). Global wide, 12,000 deaths occurred in 2019 in which 7000 and 5000 deaths were registered for boys as and girls, respectively. There was almost an 83% mortality rate documented in children and adolescent cancers in Latin American and Caribbean countries. Honduras, Saint Vincent and the Grenadines, Peru, Guyana, Ecuador, Nicaragua, and Venezuela were confirmed to have the highest mortality rates in both childhood and adolescent cancers. However, in American regions, the mortality rate in childhood and adolescent cancers was reduced from 5.1% to 4.0%/100,000 in 2000–2019. Leukemia, brain/nervous system

cancers, multiple myelomas, and lymphoma were found to be top the three cancers in 2019 (Portal, 2023).

8.3 List of childhood cancers

There are many criteria identified in documenting the different types of cancers in children but we have followed the GLOBOCAN 2012 and ICC-3 criteria to discuss the listed diseases. However, in GLOBOCON 2012, leukemias (ALL and acute myeloid leukemia [AML]), brain and nervous system tumors, non-Hodgkin lymphoma, kidney tumors (Wilms tumour), Hodgkin lymphoma, and other categorized forms of cancers were listed (Joko-Fru et al., 2024). The classification of childhood cancer is based on tumor morphology. The subclassification of childhood cancers is based on the site and morphology in ICC-3. The detailed list of childhood cancers is shown in Table 8.2.

8.4 Risk factors and its types

A risk factor is defined as an aspect of behavior or lifestyle, which is a habitual pattern of diet, exercise, cigarette and alcohol consumption, or a biological characteristic, which is known to be an inborn/inherited characteristic, a health-related condition, or environmental exposure with predictable effects on the risk of disease due to a specific cause, including an increased likelihood of an unfavorable outcome (Ehrhardt et al., 2023). The risk factors related to lifestyle will correlate with low physical activity, unhealthy diet, and consumption of sweetener beverages which lead to the development of noncommunicable diseases like cardiovascular diseases which include different forms of cancers (Atzendorf et al., 2018; Neuroscience et al., 2001). Low birth weight, elevated parental age, and congenital anomalies are considered to be the inherent risk factors for associating consistently with various forms of cancers in children (Belson et al., 2007). There are various types of risk factors related to the development of human diseases, and among them, the risk factors for the development of childhood cancers can be divided into demographic, environmental, parental, intrinsic, and genetic risk factors (Avagyan et al., 2023).

8.4.1 Demographic risk factors

The male-to-female ratio for childhood cancer was found to be 1.04–1.64 in neuroblastoma and germ cell tumors in patients between 0 and 19 years; precisely, age varies depending on age group during the various modes of diagnosis. In females, Wilms tumor was found to be very common and it is one of the major childhood cancers. The overall incidence of childhood cancer is high in infancy at about 240 cases

Table 8.2 Different forms of childhood cancers.

I. Leukemias, myeloproliferative diseases, and myelodysplastic diseases					
a) Lymphoid leukemias	b) Acute myeloid leukemias	c) Chronic myeloid leukemias	d) Myelodysplasia and other diseases	e) Specified/ unspecified leukemias	
II. Lymphomas and reticuloendothelial neoplasms					
a) Hodgkin lymphomas	b) Non-Hodgkin lymphomas (excluding Burkitt lymphoma)	c) Burkitt lymphoma	d) Miscellaneous lymphoreticular neoplasms	e) Unspecified lymphomas	
III. Central nervous system/miscellaneous intracranial/ Intraspinal neoplasms					
a) Ependymomas and choroid plexus tumor	b) Astrocytomas	c) Intracranial/ intraspinal embryonal tumors	d) Other gliomas	e) Other specified intracranial/ intraspinal neoplasms	f) Unspecified neoplasms
IV. Neuroblastoma and other peripheral nervous cell tumors					
a) Neuroblastoma and ganglioneuroblastoma	b) Other peripheral nervous cell tumors				
V. Retinoblastoma					
VI. Renal tumors					
a) Nephroblastoma and other nonepithelial renal tumors	b) Renal carcinomas	c) Unspecified malignant renal tumors			

VII. Hepatic tumors

a) Hepatoblastoma	b) Hepatic carcinomas	c) Unspecified malignant hepatic tumors			
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VIII. Malignant bone tumors

a) Osteosarcomas	b) Chondrosarcomas	c) Ewing tumor and related sarcomas of the bone	d) Other specified malignant bone tumors	e) Unspecified malignant bone tumors	
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XII. Other and unspecified malignant neoplasms

a) Other specified malignant tumors	b) Other unspecified malignant tumors				
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IX. Soft tissue and other extraosseous sarcomas

a) Rhabdomyosarcomas	b) Fibrosarcomas/ peripheral nerve sheath tumors/ other fibrous neoplasms	c) Kaposi sarcoma	d) Other specified soft tissue sarcomas	e) Unspecified soft tissue sarcomas	
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X. Germ cell tumors/trophoblastic tumors/neoplasms of gonads

a) Intracranial and intraspinal germ cell tumors	b) Malignant extracranial and extragonadal germ cell tumors	c) Malignant gonadal germ cell tumors	d) Gonadal carcinomas	e) Other and unspecified malignant gonadal tumors	
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(Continued)

Table 8.2 (Continued)

XI. Other malignant epithelial neoplasms/malignant melanomas					
a) Adrenocortical carcinomas	b) Thyroid carcinomas	c) Nasopharyngeal carcinomas	d) Malignant melanomas	e) Skin carcinomas	f) Other/ unspecified carcinomas
XII. Other and unspecified malignant neoplasms		a) Other specified malignant tumors		b) Other unspecified malignant tumors	

per million per annum. However, the incidence rate was decreased to 128 and 220 cases per millions between ages 5–9 and 15–19 years. Both race and ethnicity can influence the risk levels of childhood cancer, that is, in the United States America, the incidence of cancer in White children is high when compared with that in Asian, Black, and Hispanic children. The incidence rate of cancer is high in other groups in comparison with that in White children. The incidence rate of acute leukemia is 10% higher in Hispanic children in comparison with White children which is significant (Spector et al., 2015).

8.4.2 Environmental factors

Smog is considered one of the environmental risk factors, along with nutritional inadequacies, elevated body mass index, sanitation, significant restrictions to drinking water, and hygiene (Ren et al., 2022). Prior to the process of chemotherapy, implementing a high dose of ionizing radiation might be the cause of the development of numerous forms of cancers in children. Amendment in environmental exposure poses a challenge in expounding the effects among children with cancers. Prospective and case–control studies with limited or restricted sample size might not be sufficient to represent the meaningful analysis toward the disease. The demographic data on the parental history along with environmental factors which include the lifestyle habits will not be subject to recall errors and not an advantage for designing the retrospective studies (Spector et al., 2015).

8.4.3 Parental risk factors

Childhood cancer is a rare disease. Case–control studies were deemed the most efficient epidemiological study design to explore for the various risk factors throughout the time of delivery (Cole, 1980). The age of both parents is one of the risk factors for the development of congenital syndromes, which further predispose to cancers, as supported by a couple of studies conducted in United Kingdom (n=10,162 cases) (Dockerty et al., 2001) and Sweden (n=7,844 cases) (Yip et al., 2006) exclusively for leukemia. Furthermore, findings by Johnson et al. (2009) confirmed the preceding studies regarding parental age as a risk factor. The established risk factors include precise genetic syndromes, demographic features (e.g., gender and ethnicity), prenatal exposure to ionizing radiation, and high weight at birth (Johnson et al., 2009). Various studies, however, produced contradicting results, such as young maternal age at delivery, a history of miscarriage, maternal anemia, smoking, and maternal obesity, as well as the development of multiple childhood cancers. Furthermore, fetal growth is shown to be the most studied perinatal risk factor for the development of childhood cancer (Bhattacharya et al., 2014). Limited research has evaluated the involvement of medicinal

consumption during pregnancy to the development childhood cancers (e.g., leukemia and brain tumors), as well as the use of antinausea drugs, which were found to be both significant (Bunin et al., 1989; Kuijten et al., 1990; Robison et al., 1989) and nonsignificant (Birch et al., 1990; McKinney et al., 1985; Preston-Martin et al., 1982; Zack et al., 1991). The documented studies found that there is a lower risk of developing childhood cancers if the mother consumes multivitamin supplementations such as vitamins C and E during pregnancy (Keats et al., 2021; Kwan et al., 2007; Preston-Martin et al., 1996, 1998; Thorne et al., 1994). Tobacco smoking is considered to be the main cause of cancer. The studies were restricted to understanding the relation between childhood cancer and environmental exposure to tobacco of pregnant women. The global-wide studies showed all forms of associations (Chang, 2009; Liu et al., 2011; Secretan et al., 2009) and meta-analysis studies showed positive association (Keats et al., 2021; Lee et al., 2009; Liu et al., 2011; Milne et al., 2012). The stronger occurrence of pediatric cancers due to paternal smoking before conception is believed to be caused by germline variants that occur during spermatogenesis. Although the focus has primarily been on the risks due to smoking in pregnant females, the available research indicates that smoking by the father around the time of conception may also elevate the likelihood of cancer in their children (Behl et al., 2013; Hackshaw et al., 2011; Heck et al., 2016; Lee & Lupo, 2013; Wyrobek, 1993). However, no meta-analysis studies showed all forms of associations between childhood cancers and women smoking. Consumption of alcohol can cause different forms of cancers like pharynx, esophagus, larynx, oral cavity, liver, and breast cancer in females and colorectal cancers. It also results in exposure toward acetaldehyde, known to be a genotoxic compound, detoxified by aldehyde dehydrogenases, and this variant allele ALDH2*2 encodes an inactive enzyme. An increased incidence of alcohol-related cancers is associated with heterozygous carriers, who have approximately 10% enzyme activity and accumulate acetaldehyde, in contrast to persons with the common alleles (Secretan et al., 2009). Global-wide studies have confirmed all forms of association related to consumption of alcohol by pregnant women related to the development of childhood cancers (Auger et al., 2019; Green et al., 2002; Latino-Martel et al., 2010; Shu et al., 1996; Wimberly et al., 2024). Additionally, occupations involving motor vehicles, factories, paints, exposure to hydrocarbons and metal, pharmaceuticals, and manufacturing of textiles add significant risks to the development of childhood cancers such as AML and ALL. The possible mechanism by which occupational work is linked to the development of cancer in children is the genotype effects of fathers' spermatozoa, transplacental fetal exposure when parents bring toxic exposure into the home, and direct postnatal exposure to chemicals brought home during cloning (Savitz & Chen, 1990; Van Steensel-Moll et al., 1985; Vianna et al., 1984). Breastfeeding has been linked to a lower incidence of ALL, making exposure through breast milk unlikely (Belson et al., 2007); studies have also shown positive association (Amitay & Keinan-Boker, 2015; Investigators, 2001;

Küçükçongar et al., 2015; Ou Shu et al., 1999). Family history of previous fetal loss will tend to increase the risk of AML and ALL in their subsequent offsprings (Ross et al., 1997), and other parental risk factors can be the potential causes, which include a high chance of genetic mutations in the sperm of elder fathers and epigenetic changes in germ cells with increasing parental age (Dockerty et al., 2001).

8.4.4 Intrinsic risk factors

Intrinsic risk factors are defined as the characteristics which are inherent to the individual and which cannot be influenced by behavioral changes. Intrinsic risk factor is also the effect of individual physical variables such as age, flexibility, previous injury, and strength along with self-concept and personality which is described as psychological variables (Brewer & Redmond, 2016). The intrinsic risk ascends from the basal mutation rate operating in all dividing cells, which can be the nonpresence of any of the nonintrinsic factors. A couple of intrinsic factors relate to age and mental health. However, age-related decline is crucial; diseases affecting any of the systems involved in the balance and mental status also increase the risk, for example, glaucoma affecting vision and early-onset dementia affecting cognition (Clay et al., 2018). Acute lymphoid and myeloid leukemia in both low and high birth weights, neuroblastoma, tumors in CNS, Wilms tumor along with birth weight and birth size which is the main cause for the exposure of prenatal growth hormone. Advanced parental age may also be one of the reasons for the development of childhood cancer in infants. One of the documented studies has found correlation between maternal age and brain/bone tumors, lymphoma, leukemia, neuroblastoma, Wilms tumor, and soft tissue sarcoma, although paternal age was not associated with these cancers (Johnson et al., 2009; Veltman & Brunner, 2012). A correlation was found between structural birth defects and childhood cancers. Limited associations were explained through genetic pathway with majority of birth defects appearing to be sporadic (Botto et al., 2013; Carozza et al., 2012; Fisher et al., 2012; Hobbs et al., 2014).

8.4.5 Genetic risk factors

Genetic risk factor or inherited risk factors are passed from one generation to the second generation via genes, and childhood cancers are majorly caused by inherited syndromes, which is mainly because of high-penetrance germline DNA mutations, epigenetic disorders, and chromosomal aneuploidy. Pediatric adrenocortical carcinoma is known to be one of the rare cancer types (Pinto et al., 2020). Majority of the studied risk factors influence genes causing de novo mutations, which have their unique, determined place on cancer pathogenesis (Avagyan et al., 2023). The advantage of genome-wide association studies (GWAS) is in their large sample size, enabling comparison of the

frequencies between hundreds to thousands of common single nucleotide variants (SNVs). The convincing results are based completely on a high degree of positive statistical association. However, for a rare disease, a large sample size is not readily achievable and significance analysis is lacking. Based on the documented results, GWAS couldn't be designed for restricted samples such as neuroblastoma, acute lymphoid leukemia, osteo/Ewing sarcoma, and Wilms tumor, all which have confirmed numerous variants associated with each disease (Spector et al., 2015). Using Mendelian approach via linkage analysis, the relation between cancer and the initial gene screening was conducted and further numerous childhood tumors were recorded with cancer predisposition syndromes in dominant and recessive inheritances because of modifications in repaired DNA genes (Table 8.3). Published reports on childhood cancer have confirmed a major frequency in germline alteration (Alonso-Luna et al., 2023).

8.4.6 Genetics of cancer

The accumulation of multiple genetic amendments in a cell can lead to the development cancers, and these changes can lead to abnormal cell proliferation and clonal expansion, which is further spreads to other tissues. Genetic changes in the majority of cases which indorse tumorigenesis ensue in somatic cells and do not involve germline mutations. Many genes were enrolled toward tumor development, and furthermore, it was classified as (1) tumor suppressor genes, (2) proto-oncogenes, and (3) genes tangled in genome instability. The role of the tumor suppressor gene is to control the cell proliferation, which further inhibits the progression of the cell cycle or induces apoptosis (Vogelstein & Kinzler, 2004).

8.5 Genomic studies in childhood cancers

Cancer exists due to alteration in genetics (Greaves et al., 2003). Despite the relaity of childhood cancers, there is still a poor understanding of their etiology. The aggregation of nationwide case–control studies and facilitation of genetic advancement technology have resulted in an enhanced comprehension of the aspects linked to childhood cancers. Prenatal and perinatal exposures can be the contributing factors for the development of cancers in children in their initial life (Bhattacharya et al., 2014). In addition, the contributing factors for the development of pediatric cancers could be common genetic variants, birth weight, and birth defects (Ndlovu et al., 2022). Environmental factor is also found to be a link for the development of pediatric cancer, although it was restricted and tough to identify the causal association because of bias in their specific epidemiological studies (Ndlovu et al., 2022). Burkitt lymphoma and retinoblastoma are childhood cancers which have signs and symptoms that are easily recognizable, while acute leukemia and astrocytoma are found to have nonspecific symptoms (Magrath et al., 2013).

Table 8.3 List of childhood tumors and its association with predisposition syndromes.

Tumor(s)	Predisposition syndromes	Related gene amendments
Leukemia	Fanconi anemia, neurofibromatosis type-1, ataxia telangiectasia, Li-Fraumeni, and Nijmegen hereditary retinoblastoma	FANCA, FANCB, BRCA, NF1, ATM, TP53 and RB1
Lymphoma	Neurofibromatosis type-1, ataxia telangiectasia, Wiskott-Adrich, Bloom, Nijmegen, Li-Fraumeni, and Beckwith-Wiedmann	NF1, ATM, WASP, BLM, NBS1, TP53, BRCA (CDKN1C, H19, IGF2, and KCNQ10T1)
Central nervous system tumors	Rhabdoid tumor, Li-Fraumeni, Noonan, multiple endocrine neoplasia type 1, neurofibromatosis type 2, ATM, Beckwith-Wiedmann, Li-Fraumeni, Von Hippel Landau, and DICER1	SMARCB1, TP53, RAS, MEN1, NF2, ATM, (CDKN1C, H19, IGF2, and KCNQ10T1), TP53, VHL, DICER1
Neuroblastoma	Neurofibromatosis type 1, Li-Fraumeni, Beckwith-Wiedmann, and Noonan	NF1, TP53 (CDKN1C, H19, IGF2, KCNQ10T1) and RB1
Retinoblastoma	Hereditary retinoblastoma	RB1
Renal tumors	Beckwith-Wiedmann, Li-Fraumeni, and Denys-Drash	(CDKN1C, H19, IGF2, KCNQ10T1), TP53 and WT1
Hepatic tumors	Beckwith-Wiedmann, Li-Fraumeni, and Fanconi anemia	(CDKN1C, H19, IGF2, KCNQ10T1), TP53, FANCA, FANCB and BRCA
Malignant bone tumors	Li-Fraumeni, hereditary retinoblastoma, and Beckwith-Wiedmann	TP53, RB1 (CDKN1C, H19, IGF2 and KCNQ10T1)
Soft tissue sarcoma	Li-Fraumeni, hereditary retinoblastoma, neurofibromatosis type 1, DICER1, and Beckwith-Wiedmann	TP53, RB1, NF1, DICER1
Germ cell tumors	Denys-Drash, Cowden Noonan, and Peutz-Jeghers	WT1, PTEN, RAS, STK11
Epithelial tumors	Li-Fraumeni, Beckwith-Wiedmann, Xeroderma pigmentosa, and familial melanoma	TP53, (CDKN1C, H19, IGF2, KCNQ10T1), (XPA-XPG y ERCC5), CDKN2A

Chromosome translocation is involved majorly in multiple childhood leukemias which is known to originate during fetal hematopoiesis based on the studies conducted on monozygotic twins concordant for leukemia (Greaves et al., 2003). Gröbner et al.

(2018) conducted a cohort study that examined 961 tumors from children, adolescents, and young adults, recording 24 different forms of cancers. Genetic screening was performed on all samples, and genetic changes in 149 suspected cancer driver genes classified the tumors into two types: minor mutations and structural/copy number variants (which correspond to germline variants). Gröbner et al. (2018) found that 7%–8% of children had a clear predisposing germline disparity and ~50% of pediatric neoplasms have a potentially druggable event. This information is important for designing future clinical studies.

A previous study documented that 8.5% of young adults (who were in twenties) were carriers of germline mutations in the *TP53*, *APC*, *BRCA2*, *NF1*, *PMS2*, *RB1* and *RUXN1* genes. This rate was significantly higher than the 1% observed in the control population. Conspicuously, 40% of carriers in the control population had a family history of cancers (Zhang et al., 2015). Fourteen percent was found to be the highest percentage of germline mutations in childhood cancers, which were the carriers of modifications in cancer predisposition genes, and it was confirmed through whole exome sequencing (WES) analysis (Oberge et al., 2016). Studies by Ballinger et al. (2016) have confirmed some of the limitations of a large-sample size study such as the following: (i) all tumors were not included, (ii) inheritance and origin of alterations couldn't be established because sequencing analysis included only children, and (iii) environmental factors were not discussed. Based on these limitations, future studies were recommended to construct huge family-based germline sequencing studies in order to establish the origin of the alterations, that is, parents/de novo, evaluation of penetrance, and gene–environment interaction analysis (Ballinger et al., 2016). Among complex diseases, GWAS often establishes an association through the analysis of SNPs or SNVs. The novel variant trait association can be identified by screening the SNPs in identifying new biological mechanisms and explaining the ethnic variation role. However, scrutinizing of novel SNPs was restricted (Tam et al., 2019).

GWAS studies conclude as multiple SNPs could be having cumulative risk of cancer and also act as a genetic modifier in addition with disease-causing variants, which further results in oligogenic and polygenic diseases with variations in penetrance/expressivity/quantitative genetic epistasis (Rahit & Tarailo-Graovac, 2020). Because the nominal price was applicable for SGS studies, the number was increased for designing GWAS studies since a decade throughout the globe, and with 3–4-digit SNPs, either intermediate or low penetrance was documented in association with cancer. When GWAS compared childhood and younger and adult cancers, the odds ratio analysis confirmed that adult cancers were significantly associated with an elevated proportion of populations with attributable risk (Raynor et al., 2013). One of the main reasons for lacking GWAS in childhood cancer is lower sample size—more importantly, childhood cancer epitomizes only 1% of all cancer cases. The strength of GWAS and meta-analysis studies will always aid in population genetic studies (Evangelou & Ioannidis, 2013).

Epidemiological studies were found to have an important role in studying and confirming the common disease–common variants hypothesis via GWAS which can identify the minor risk of human diseases and also evaluate the interaction analysis with both lifestyle and environmental factors associated with minimum- or moderate-penetrance SNPs (Alonso-Luna et al., 2023).

A large scale of NGS data can be beneficial in both the diagnosis and prognosis of various forms of childhood cancers. Genomic data were obtained mainly from human genome project, as well as from a second-generation sequencing (SGS) analysis including next-generation and exome sequencing techniques with other patterns of SGS analysis along with the advent of omic sciences, which has become very common since a couple of decades. The classification of genomics in various forms of mechanisms, that is, molecular and genomics of pediatric oncology, confirms the landscape of genomic cancer, which is not common in childhood cancers. When compared to the etiology and tumorigenic pathways of typical adult cancers, many are remarkably different. The advancement of SGS techniques has provided an advantage toward identifying the 10% of the pathogenic germline variants in the pediatric population who were diagnosed with the cancers. Presently, above hundred genes were confirmed for cancer susceptibility and were found to be pathogenic germline variants, that is, loss of functional variants, that is, gene encodes DNA double-stranded repaired proteins (Cipri et al., 2022). The approach of NGS is vital in pediatric cancer for identifying phenotypic variabilities (Boccuto et al., 2022). The combined NGS data, that is, whole genome sequencing (WGS), WES, and RNA sequencing (RNA-seq), can lead to the improvement of cancer disease treatment in children. The use of radiation or medications that raise the risk of a second malignancy should be minimized in treatment options for children with germline changes in cancer predisposition genes, especially those involving chromosomal instability or DNA repair sites (Cipri et al., 2022). The WES analysis studied was found to be valuable tool in identifying the somatic and germline variants present in T-cell for both acute lymphoid/myeloid leukemia and Wilms tumors in pediatric cancers (Bolouri et al., 2018; Gadd et al., 2017; Liu et al., 2017). WES analysis identified the variants present in CTNNB1 and NFE2L2 genes present in hepatic cell carcinoma (Vilarinho et al., 2014), as well as variants present in the SMARCB1 gene among 35 patients diagnosed with Rhabdoid tumor (Lee et al., 2012). The NGS studies were also useful in identifying the genetic variants that exist either as somatic or as germline (Parsons et al., 2016; Zhang et al., 2015).

RNA-seq analysis is a power tool used for detecting gene fusion among the pathogenic variants (e.g., *NUTM1* gene) (Boer et al., 2021). Using Ricarte-Filho et al. (2013) study technique confirmed two different types of fusions in the MAPK pathway in thyroid cancers: (i) somatic rearrangement fusions present in the *ETV6* and *NTRK1* genes, and an *AGK* fusion in the *BRAF* gene. Fusion genes can play a role with sarcomas and leukemia in carcinogenesis using the expression of oncogenes. For other

childhood cancers, fusion genes have no role. However, oncogenic effects were present and identified for osteosarcoma and medulloblastoma which are further proximal to the enhancer elements (Chen et al., 2014; Northcott et al., 2014). LaHaye et al. (2021). Studies confirmed RBPMS-MET, BCAN-NTRK1, and TRIM22-BRAF gene fusions. Gröbner et al. (2018) have confirmed MSH6 and PMS2 genes using WGS and WES studies among 547 and 414 samples of pediatric cancer samples. Integrated investigations combining WGS and RNA-seq across various tumors have proven to be valuable for cancer diagnosis. These genetic data can be used to diagnose the prognosis of childhood cancer during their course of treatment (Cavalli et al., 2018; Clarke et al., 2020; Mack et al., 2018; Parker et al., 2014; Robinson et al., 2018; Taylor et al., 2012).

Additionally, modifications of germline variants in cancer treatments are useful in understanding the cancer etiology, standardizing the initial therapy, and developing surveillance protocols. The germline variants in different forms of cancers, such as the *DICER1* gene in pleuropulmonary blastoma, *RB1* for retinoblastoma, *TP53* for adrenocortical carcinoma, and *SMARCB1/SMARCA4* for rhabdoid tumor/small cell ovarian cancers, are well-documented (Salas et al., 2017). The screening of germline variants among children diagnosed with atypical teratoid/rhabdoid tumor in *SMARCB1* gene is essential for early risk assessment. The similar variant was documented in brain/kidney tumors, predisposition factors for the development of rhabdoid tumors (Biegel et al., 1999; Eaton et al., 2011; Foulkes et al., 2017). Additionally, from global studies, different germline variants such as *POLR2A*, *DICER1*, *WT1*, *CHEK2*, *PALB2*, *TRIM28*, *REST*, and *CTNNB1* were documented in the published articles (Boni et al., 2021; Gadd et al., 2017; Huff, 1998; Mahamdallie et al., 2015, 2019; Palcuict et al., 2016; Paparella et al., 2022). These reported studies indicate that the genome plays a crucial part in comprehending the analysis of cancer propensity. It is vital to be aware of this while selecting the appropriate disease for diagnosing tumors and executing genome analysis investigations.

Genomic studies have determined the prevalence of mutations in genes and the development of risk stratification, leading to the identification of the most appropriate medicines. The WES studies confirmed low-grade NF1 gliomas among biopsy samples of cohort studies (D'Angelo et al., 2019). A high frequency of variants was present in *TP53*, *IDH1*, and *ATRX* genes among children with glioma (Ceccarelli et al., 2016; Chatsirisupachai et al., 2021; Lee et al., 2021; Li et al., 2022). Genomic studies were used to develop personalized therapies guided by alterations, which has shown exceptional results with targeted drugs in treating anaplastic large cell lymphoma and myofibroblastic tumors (Lourenço et al., 2012). A pan-cancer genome study was carried out in 1699 for pediatric leukemia and solid tumors and confirmed somatic variants exist in *TP53*, *KRAS*, *NRAS*, and *CDKN2A* levels. WGS studies identified 72% of tumors in neuroblastoma which was further compared to one driver gene in ~26% of the samples, while WGS analysis confirmed 11% of samples confirm the existence of

chromothripsis (Ma et al., 2018). A genomic study of non-Down Syndrome acute megakaryoblastic leukemia was conducted on 75 pediatric and 24 adult cancer patients using WES and RNA-seq. The results suggest that the disease in pediatric patients can be characterized as a heterogeneous malignancy (De Rooij et al., 2017). Studies by Zhang et al. have confirmed that 8%–10% of pediatric cancers are associated with germline mutations which lead to the progression of cancer (Zhang et al., 2015).

8.6 Is there any diagnosis with genomics in pediatric cancers

In the last few decades, we have committed to the advancements in the field of molecular biology which were helping us understand the molecular pathways in tumors. This advent of molecular genomics is providing unrivalled insights into the molecular landscape of pediatric cancers or tumors. The identification of molecular markers is essential for the appropriate diagnosis, risk stratification, disease subtyping, surveillance, and tailored cancer management. The most common childhood tumor retinoblastoma is caused mostly by mutations in *RB1* gene and to an extent by MYCN amplifications. If the molecular variants are identified initially, childhood cancers can be treated in a better way with a tailored treatment approach. This genome testing will also be obliging toward other age group-related subjects diagnosed with cancers (Stadler et al., 2014). Limited cancers were documented for specific causes, that is, smoking in lung cancer, sunlight in skin cancer, human papilloma virus infection in cervical cancer, *Helicobacter pylori* infection in gastric cancer, and viral hepatitis in hepatocellular carcinoma, although studies on prostate and colorectal cancers were restricted to understand completely (Knowles & Selby, 2005; Wu et al., 2018).

In terms of genome studies in pediatric cancers, fusion gene or gene fusion has an impact in diagnosis, oncogenesis, therapeutic targets, and biomarkers. Childhood cancers have a low mutational rate, and their etiology can be explained by chromosomal rearrangements, which is an alternate mechanism of oncology and further supports the potential role of fusion genes in the development of pediatric cancer (Filbin & Monje, 2019). A pair of separate genes that are linked, transcribed, and translated as a single unit is defined as gene fusion or fusion gene. In nature, they have evolved over time, for example, a fusion protein that performs both steps of a metabolic process is born when two genes' products combine. Gene fusion can be induced in vitro using the rDNA approach, as well as in vivo, which is when it happens naturally as a result of genetic modification (Maloy & Hughes, 2013). The history of fusion genes begins in 1962, when Nowell (1962) defined the first fusion gene in chronic myeloid leukemia (CML). Later, in 1973, Rowley (1973) highlighted the existence of reciprocal translocation between the long arms of chromosomes 9 and 22, that is, t(9q; 22q). Gene fusion or chromosomal rearrangements are important aspects of somatic variants in cancers, and Philadelphia chromosome (t9:22) was the first cancer-associated chromosome

rearrangement. ABL1 and BCR are a couple of genes connected with gene fusion in different forms of leukemia. Additional examples exist in hematological malignancies and sarcomas as follows: MLL fusions, RUNX1-RUNX1T1, PML-RAR α , EWSR1-FLI1-FL11, and EVT6-NTRK3. The first fusion gene present in solid tumor was CTNNB1-PLAG1 (Taniue & Akimitsu, 2021). The molecular mechanisms underlying fusion genes are classified as insertion, deletion, translocation, and inversion. The physiological implications differ depending on the affection of the coding sequences. The majority of the fusion genes in pediatric cancer occur in leukemia, which is quite prevalent. There are two categories of fusion genes present in pediatric solid tumors, that is, diagnostic markers and new therapeutic targets (Sweet-Cordero & Biegel, 2019). Ewing sarcoma comes under the first group, and it has a reciprocal translocation [t(11;22)(q24;q12)] which leads the fusion between EWS and FL11 genes (Delattre et al., 1992). Another translocation present in Ewing sarcoma is PAX3/7-FKHR fusions, which result from the translocation t(2;13)(q35;q14) or, less frequently, t(1;13)(p36;p14). These fusions produce the PAX3-FOXO1 (55%–70%) or PAX7-FOXO1 (10%–22%) fusion genes, respectively, in alveolar rhabdomyosarcomas (de Souza et al., 2014). The target therapeutic was another form of fusion genes which incorporates ASPSCR1-TFE3, FGFR1-TACC1, FGFR3-TACC3, KIAA1549-BRAF, NPM-ALK, and NTRK fusions (Amatu et al., 2016). Liu et al. (1739) conducted oncogenic fusion studies on 5190 pediatric cancers in US children, and the findings validated the several parameters that shape the formation of oncogenic fusions, including translation frame, protein domain, splicing, and gene length. In addition, they uncovered RUNX1-RUNX1T1, TCF3-PBX1, CBFA2T3-GLIS2, and KMT2A-AFDN oncogenic fusions, as well as neo splice sites in 18 oncogenic fusion gene pairs and demonstrated that such splice sites give therapeutic vulnerability for etiology-based genome editing (Liu et al., 1739).

8.7 Conclusion

Genomics is a branch of molecular biology that deals with every aspect of a gene, including structure, function, evolution, and mapping. The role of genomics is to extend the nucleotides contained in double-stranded DNA and examine them in depth using different methods of sequencing analysis, which is used to detect mutations associated with cancers that are present in the patient. Oncologists can prescribe specific medications based on the presence of cancer mutations in children. Last but not least, this chapter concludes that the cancer genome in the pediatric population is significant in detecting the disease and contributing to precise medical treatments. Finally, identification of genomic alterations has provided the opportunity to derive new targeted strategies for the clinical management of adult and pediatric cancers which is referred to as precision medicine which further includes the noninvasive detection, risk

assessment, molecular diagnosis, and targeted therapies (Cipri et al., 2022). Furthermore, this genome testing will help in planning preimplementation and parental genetic testing if required to prevent the morbidity caused by these types of cancers in children.

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